

Minor Project Synopsis Report

ADVANCED QUADRUPE SURVEILLANCE ROBOT

Project Category: IoT Based

Projexa Team Id-26E2099

BACHELORS OF TECHNOLOGY
in
CSE with Specialization Cyber Security

to

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ABSTRACT

Monitoring hazardous, confined, or disaster-prone environments such as research laboratories, industrial warehouses, chemical plants, and disaster-affected sites poses significant risks to human personnel. These environments are often characterized by uneven terrain, debris, staircases, narrow passages, toxic substances, and unstable structures, which severely limit the effectiveness of traditional wheeled robotic systems and make direct human intervention unsafe or impractical. While quadruped robots are inherently well suited for traversing such challenging terrains due to their superior balance, adaptability, and legged locomotion, many existing quadruped platforms lack robust Internet of Things (IoT) integration, advanced environmental sensing capabilities, secure real-time data streaming mechanisms, and reliable support for long-duration autonomous missions.

This project proposes the design and development of an advanced IoT-enabled quadruped robotic surveillance platform capable of autonomous navigation and intelligent decision-making in hazardous environments. The proposed system integrates high-resolution visual cameras along with thermal imaging sensors to capture and stream real-time video feeds and heat-mapping data to remote monitoring and research teams. By employing multi-sensor fusion techniques involving LiDAR, ultrasonic sensors, and an Inertial Measurement Unit (IMU), the robot achieves accurate environmental perception, precise mapping, stable locomotion, and intelligent path-following even in complex and dynamic surroundings.

Furthermore, the system incorporates a robust data management framework wherein all surveillance data—including real-time positional tracking, detected environmental or thermal anomalies, mission logs, and battery health metrics—are securely transmitted and stored in a MongoDB database. This data is made accessible through an intuitive web-based dashboard that enables remote monitoring, visualization, and collaborative analysis by geographically distributed research teams. Overall, the proposed quadruped robotic platform significantly enhances operational safety, situational awareness, and research efficiency, while enabling reliable long-duration autonomous surveillance missions in environments that are otherwise dangerous or inaccessible to humans.

KEYWORDS: *Quadruped Robot, IoT, Surveillance, Thermal Imaging, Sensor Fusion, MongoDB, Autonomous Navigation*

1. INTRODUCTION

Surveillance and inspection of hazardous environments have become increasingly important across a wide range of industries, including disaster management, research laboratories, industrial warehouses, chemical plants, and large-scale manufacturing facilities. These environments often present severe operational challenges such as unstable and uneven terrain, staircases, scattered debris, confined pathways, toxic or flammable gases, low visibility, and high-temperature zones, all of which make direct human access dangerous, time-consuming, and inefficient. In such conditions, prolonged human presence can lead to serious safety risks, including injury, exposure to harmful substances, and operational delays. Conventional wheeled robotic systems are also significantly limited in these scenarios due to their poor adaptability to uneven terrain, inability to climb stairs, and restricted mobility in cluttered or confined spaces.

Quadruped robots, inspired by biological locomotion observed in animals, provide a highly effective alternative due to their superior balance, stability, and adaptability in rough and unpredictable environments. Their legged structure allows them to step over obstacles, climb stairs, and maintain stability on uneven surfaces where wheeled robots typically fail. However, despite these mechanical advantages, many existing quadruped robotic platforms primarily emphasize locomotion, gait control, and structural design, while offering limited support for comprehensive Internet of Things (IoT) integration, real-time environmental monitoring, intelligent decision-making, and secure handling of mission-critical data. In modern research, industrial inspection, and disaster-response scenarios, it is no longer sufficient for a robotic system to merely navigate difficult terrain; it must also be capable of collecting, processing, storing, and transmitting large volumes of multi-sensor data in real time to support informed decision-making. This project addresses these limitations by developing an advanced IoT-enabled quadruped surveillance robot that integrates robust mobility with intelligent sensing, autonomous navigation, real-time visual and thermal data streaming, and cloud-based data management. The proposed system is designed to continuously gather environmental data, detect anomalies, and provide actionable insights to remote research and monitoring teams through secure communication channels. By enabling real-time situational awareness and remote access to critical information, the system significantly reduces the need for direct human exposure to hazardous environments while enhancing operational efficiency, safety, and reliability in complex and high-risk scenarios.

2. MOTIVATION

The motivation for this project arises from the growing demand for safe, reliable, and autonomous monitoring solutions in hazardous, confined, and high-risk environments. Situations such as natural disasters, industrial accidents, chemical leaks, structural collapses, and research operations in unsafe zones often require continuous surveillance, search-and-rescue assistance, environmental assessment, and real-time situational awareness. Performing these tasks manually exposes human responders to severe risks, including physical injury, toxic exposure, extreme temperatures, and structural instability, thereby highlighting the urgent need for robotic systems capable of operating safely in such conditions.

Although aerial drones provide a useful means of surveillance from above, they are constrained by limited battery life, reduced stability in enclosed or indoor environments, susceptibility to environmental disturbances, and an inability to closely inspect ground-level obstacles or carry meaningful payloads for extended durations. Similarly, conventional wheeled robots are restricted by their inability to navigate staircases, rubble, narrow passages, and uneven or slippery surfaces, making them ineffective in many real-world disaster and industrial scenarios. Quadruped robots address these mobility challenges through their legged design, which offers enhanced balance, adaptability, and terrain traversal capabilities. However, despite their mechanical advantages, quadruped robots remain underutilized due to insufficient integration of advanced sensing technologies, inefficient data management frameworks, limited real-time communication capabilities, and lack of seamless remote accessibility.

This project is motivated by the need to bridge the existing gap between advanced robotic mobility and intelligent Internet of Things (IoT)-based data handling and communication systems. By integrating real-time high-resolution video streaming, thermal imaging for heat signature detection, autonomous navigation through multi-sensor fusion, and secure cloud-based data storage, the proposed system enables extended-duration autonomous missions and effective collaboration among geographically distributed research teams. The ability to remotely monitor, analyze, and respond to critical environmental data significantly enhances operational safety, improves decision-making efficiency, and reduces reliance on direct human intervention, making the proposed quadruped robotic platform a valuable solution for high-risk surveillance and inspection applications.

3. LITERATURE REVIEW

AUTONOMOUS QUADRUPEL ROBOTS FOR ROUGH TERRAIN:

Recent research in the field of legged robotics highlights the effectiveness of quadruped robots in navigating uneven, cluttered, and unpredictable terrains commonly found in disaster zones and industrial environments. Multiple studies demonstrate that quadruped robots significantly outperform traditional wheeled robotic systems in tasks such as stair climbing, rubble traversal, slope handling, and obstacle negotiation. This advantage is primarily attributed to their adaptive gait control, distributed leg coordination, and advanced balance mechanisms that allow continuous adjustment to changing ground conditions. Additionally, quadruped robots exhibit improved stability and fault tolerance, enabling continued operation even when one or more legs encounter slippage or obstruction, making them highly suitable for hazardous and confined environments.

IoT-BASED SURVEILLANCE SYSTEMS:

Internet of Things (IoT)-enabled surveillance systems have been widely adopted in applications such as smart cities, industrial monitoring, infrastructure security, and environmental observation. These systems primarily focus on real-time data acquisition, cloud-based storage, remote visualization, and centralized control through web or mobile interfaces. IoT architectures enable seamless connectivity between sensors, processing units, and monitoring platforms, facilitating efficient data sharing and analysis. However, most existing IoT surveillance systems are static in nature, relying on fixed cameras or sensor nodes that lack mobility and adaptability. As a result, their effectiveness is limited in hazardous terrains where dynamic navigation and close-range inspection are required, thereby restricting their applicability in disaster-response and high-risk environments.

THERMAL IMAGING FOR SEARCH AND RESCUE:

Thermal imaging technology plays a crucial role in search-and-rescue operations by enabling the detection of human presence through heat signatures, especially in low-visibility conditions such as smoke, darkness, fog, or debris-covered areas. Numerous studies have shown that thermal sensors significantly enhance situational awareness by identifying victims, overheated equipment, or abnormal temperature patterns that are not visible to standard cameras. When integrated with mobile robotic platforms, thermal imaging enables continuous scanning of environments while maintaining a safe distance from hazards. This integration greatly improves victim detection accuracy, reduces response time, and supports efficient anomaly identification in disaster and emergency scenarios.

MULTI-SENSOR FUSION FOR NAVIGATION:

Research in autonomous robotics strongly emphasizes the importance of multi-sensor fusion techniques for accurate mapping, localization, and navigation in complex environments. Sensor fusion combines data from multiple sources

such as LiDAR, ultrasonic sensors, and Inertial Measurement Units (IMUs) to compensate for the limitations of individual sensors. LiDAR provides precise distance measurements and environmental mapping, ultrasonic sensors assist in short-range obstacle detection, and IMUs contribute orientation and stability data. By fusing these sensor inputs, robotic systems achieve improved navigation reliability, reduced uncertainty, and enhanced robustness in dynamic and cluttered environments. Multi-sensor fusion is therefore a critical component in enabling autonomous path planning and stable locomotion for quadruped robots operating in hazardous conditions.

S. No.	Title / Area of Study	Authors / Source	Methodology / Technology Used	Key Findings	Limitations Identified
1	Autonomous Quadruped Robots for Rough Terrain Navigation	IEEE Robotics Research Papers	Quadruped locomotion models, adaptive gait control, balance algorithms	Quadruped robots outperform wheeled robots in uneven terrain, stair climbing, and rubble traversal due to superior stability and adaptability	Focused mainly on locomotion; limited integration of IoT, real-time data streaming, and cloud-based monitoring
2	IoT-Based Surveillance Systems	Various IoT and Smart City Studies	IoT sensors, cloud storage, real-time data transmission, dashboards	Enables real-time monitoring, remote access, and centralized data visualization	Mostly static systems; lack mobility and adaptability in hazardous or confined environments
3	Thermal Imaging for Search and Rescue Operations	Disaster Robotics Research	Thermal cameras, heat signature detection, image processing	Effective in detecting human presence and anomalies in low-visibility environments such as smoke and darkness	Often deployed on stationary systems or drones with limited operational duration
4	Multi-Sensor Fusion for Robotic Navigation	Robotics and Automation Journals	LiDAR, ultrasonic sensors, IMU, sensor fusion algorithms	Improves mapping accuracy, localization, and navigation reliability by compensating sensor limitations	Computational complexity and integration challenges in real-time systems
5	Mobile Robots in Disaster Response	International Journal of Advanced Robotics	Mobile robotic platforms, obstacle avoidance, remote monitoring	Reduces human risk and improves situational awareness in disaster scenarios	Limited endurance, insufficient data management, and lack of unified sensing and control architecture

4. GAP ANALYSIS

Existing research efforts have successfully addressed individual components related to autonomous systems, such as quadruped robot locomotion, IoT-based surveillance frameworks, thermal imaging for anomaly detection, and sensor-based navigation techniques. Numerous studies demonstrate the mechanical effectiveness of quadruped robots in navigating uneven terrain, while separate research highlights the advantages of IoT surveillance systems for real-time data acquisition and remote monitoring. Similarly, thermal imaging has been extensively explored for detecting human presence and abnormal heat signatures in low-visibility environments. However, despite these advancements, the development of a comprehensive and fully integrated system that combines quadruped mobility with real-time IoT-based video streaming, thermal sensing, autonomous navigation, and secure end-to-end data management remains limited.

Most existing quadruped robotic platforms primarily focus on locomotion, balance control, and mechanical design, offering minimal support for scalable cloud-based data storage, real-time analytics, and collaborative dashboards required for long-term research and mission analysis. On the other hand, many IoT-based surveillance systems are static in nature, relying on fixed cameras or sensor nodes that are unsuitable for deployment in hazardous, confined, or dynamically changing terrains. These systems also lack the mobility and adaptability required for effective inspection and monitoring in disaster-prone environments.

This project addresses the identified research gap by proposing a unified platform that integrates advanced quadruped mobility with intelligent multi-sensor sensing, real-time video and thermal data streaming, autonomous navigation, and robust MongoDB-based data storage. By enabling secure data management and providing a web-based collaborative dashboard for remote access and analysis, the proposed system bridges the gap between mobile robotic surveillance and scalable IoT infrastructure, offering a practical and research-oriented solution for hazardous environment monitoring.

5. PROBLEM STATEMENT

Monitoring hazardous and confined environments poses significant risks to human operators and challenges to conventional robotic systems. Wheeled robots fail on uneven terrain, while existing quadruped robots often lack advanced sensing, real-time data streaming, and secure long-duration data management.

There is a need for an autonomous robotic system capable of navigating rough terrain, detecting environmental anomalies, streaming real-time visual and thermal data, and securely storing mission-critical information for remote analysis and collaboration. The absence of such an integrated solution limits the effectiveness of surveillance, research, and disaster-response operations.

6. OBJECTIVES

1. To design and develop an autonomous quadruped robotic platform capable of navigating complex and uneven terrains, including staircases, debris-filled areas, and confined spaces, while maintaining stability, balance, and operational reliability in hazardous environments.
2. To integrate high-resolution visual cameras and thermal imaging sensors into the quadruped platform for continuous real-time surveillance, enabling effective detection of human presence, abnormal heat signatures, and environmental anomalies under low-visibility and high-risk conditions.
3. To implement advanced multi-sensor fusion techniques using LiDAR, ultrasonic sensors, and an Inertial Measurement Unit (IMU) to achieve accurate environmental mapping, obstacle detection, localization, and intelligent path planning for autonomous navigation.
4. To enable secure and reliable real-time data streaming of visual feeds, thermal imagery, and positional information using Internet of Things (IoT) technologies, ensuring low-latency communication between the robotic system and remote monitoring stations.
5. To design and deploy a robust data management system that securely stores surveillance data, mission logs, detected anomalies, and battery health metrics in a MongoDB database, facilitating long-term storage, retrieval, and analytical processing.
6. To develop an intuitive web-based dashboard interface that provides remote access to live video streams, sensor data, alerts, and historical records, thereby enabling collaborative monitoring, analysis, and decision-making by geographically distributed research teams.
7. To ensure reliable and efficient operation of the robotic system for extended autonomous missions of approximately 25–30 minutes while carrying research payloads, optimizing power management, system performance, and operational endurance.

7. Tools/Technologies Used

1. Microcontroller:

The microcontroller serves as the central processing and control unit of the quadruped robotic system. It is responsible for integrating inputs from various sensors, executing control algorithms, managing motor operations, and handling communication with external systems. The microcontroller coordinates real-time decision-making, ensuring synchronized locomotion, sensor data processing, and system stability during autonomous operation.

2. Servo Motors and Brushless Motors:

Servo motors and brushless DC motors are employed to achieve precise control over the robot's leg movements and joint articulation. These motors enable accurate gait generation, smooth locomotion, and effective load distribution while carrying research payloads. Brushless motors provide high efficiency and reliability, making them suitable for extended-duration missions, while servo motors allow fine-grained positional control of individual joints.

3. LiDAR and Ultrasonic Sensors:

LiDAR and ultrasonic sensors are used for obstacle detection, distance measurement, and real-time environment mapping. LiDAR provides high-resolution spatial data for accurate mapping and navigation, whereas ultrasonic sensors assist in short-range obstacle detection, especially in confined or cluttered spaces. Together, they enhance the robot's perception of its surroundings and support autonomous path planning.

4. Thermal Sensor:

The thermal sensor enables the detection of heat signatures emitted by humans, animals, or operating equipment. This capability is particularly valuable in low-visibility conditions such as smoke, darkness, or debris-covered environments. Thermal imaging assists in identifying potential survivors, overheated machinery, or abnormal temperature variations indicative of hazardous conditions.

5. IMU (Inertial Measurement Unit):

The IMU provides real-time orientation, acceleration, and angular velocity data, which is essential for maintaining balance, posture, and stability during locomotion. By continuously monitoring the robot's movement and orientation, the IMU supports adaptive gait control and prevents instability while traversing uneven terrain.

6. WiFi Connectivity:

WiFi communication enables secure real-time data transmission between the quadruped robot and remote monitoring systems. It supports live video streaming, thermal data transfer, positional updates, and system health reporting, allowing operators to remotely monitor and control the robot during missions.

7. MongoDB Database:

MongoDB is used as the primary data storage solution for managing structured and unstructured surveillance data. It stores mission logs, sensor readings, detected anomalies, video metadata, and battery performance metrics. The scalable and flexible nature of MongoDB supports efficient data retrieval and long-term analysis for research and evaluation purposes.

8. Web-Based Dashboard:

The web-based dashboard provides an intuitive and user-friendly interface for live monitoring, data visualization, and collaborative analysis. It allows authorized users to access real-time video feeds, sensor data, alerts, and historical records remotely, facilitating informed decision-making and collaborative research across distributed teams.

8.METHODOLOGY

The proposed quadruped robotic system is designed to operate autonomously using either a predefined patrol route or a dynamically generated navigation path, depending on mission requirements and real-time environmental conditions. Path selection and movement decisions are continuously influenced by sensor inputs to ensure safe and efficient traversal through hazardous and confined environments. LiDAR and ultrasonic sensors actively scan the surroundings to detect obstacles, measure distances, and identify navigable paths, while data from the Inertial Measurement Unit (IMU) is used to maintain balance, posture, and stability during locomotion across uneven terrain and staircases.

To achieve accurate environmental perception and reliable navigation, multi-sensor fusion algorithms are employed to combine data from LiDAR, ultrasonic sensors, and the IMU. This fused data enables the generation of precise environmental maps, supports real-time localization, and guides intelligent path-following and obstacle-avoidance decisions. By compensating for the limitations of individual sensors, the sensor fusion approach improves navigation robustness and operational reliability in dynamic and cluttered environments.

Simultaneously, high-resolution visual cameras and thermal imaging sensors capture continuous live feeds of the surrounding environment. These data streams are securely transmitted over WiFi to a remote server, enabling real-time surveillance and situational awareness for remote operators and research teams. Thermal data is analyzed to detect heat anomalies, human presence, or overheated equipment, while visual feeds provide contextual information for accurate assessment.

All mission-related data, including real-time positional tracking, detected thermal anomalies, environmental sensor readings, system health parameters, and battery performance metrics, are securely stored in a MongoDB database. This centralized data storage facilitates efficient retrieval, long-term analysis, and historical comparison. A web-based dashboard interfaces with the database to present live video feeds, alerts, system status, and detailed mission logs, thereby enabling effective monitoring, analysis, and collaborative decision-making during and after mission execution.

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